High Definition Oncology

Statistical analysis plan for manuscript revision and evaluation in new participants

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Document control and approval

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Reading guide

Sections 1 to 7 define scope, data, clocks, populations, outcomes and the frozen system. Section 8 specifies the reconstruction of the submitted analyses. Sections 9 to 19 specify the revision analyses and evaluation. Sections 20 to 23 define inference, reporting, execution and amendments. Appendix A maps analyses to reviewer comments; Appendix B provides the algorithm freeze certificate. The separate Data provenance and freeze annex contains the source inventory and clinical adjudication register.

1 Purpose and scope

This SAP governs the revision of the HDO digital phenotyping manuscript and the evaluation of its final alarm in participants who were not used for its development. The first analytical task is to reconstruct all affected submitted results after clinical audit of the original 178-participant dataset. The reconstructed development analyses will establish the final alarm specification. That specification will then be applied without fitting, relabelling or threshold adjustment to the approved independent validation evaluation population.

The plan includes within-person digital changes, expanded plasma ctDNA analyses, LDA stability, temporal resolution, simple clinical and digital comparators, missing data, pretraining-domain assessment, tumour and treatment heterogeneity, concomitant medication analyses and a simulation of an intended clinical pathway. It also specifies internal evaluation of the development procedure. These analyses address different questions and will be reported with their appropriate evidential status.

The original feasibility SAP and the retrospective Analytical Development and Provenance Report remain separate documents. Analyses already executed before this version, including any work in progress, must be identified in the execution ledger with their available dates and data access history. The present plan prespecifies only decisions and evaluations not already informed by their results. Reporting will follow TRIPOD+AI, with explicit assessment of the development and evaluation risks identified in PROBAST+AI [1,2].

2 Study overview and analysis framework

HDO is a prospective observational, multicentre study in women with advanced breast, lung or colorectal cancer, with clinical assessment, smartphone and wearable monitoring, and longitudinal biospecimen collection. The clinical treatment origin is the first dose of systemic treatment in the advanced-disease setting. The submitted manuscript used 178 participants for passive-signal analyses and a smaller population for DCABP analyses requiring sufficient monitoring.

The digital system consists of externally pretrained VQ-VAE day types, an HDO-derived LDA representation, a fixed grouping of topics into favourable and unfavourable states, and a rolling burden alarm. In the submitted system, six topics were used, topics 0 to 2 were classified as favourable and topics 3 to 5 as unfavourable. The grouping and alarm selection were informed by outcomes in development. Unsupervised representation learning does not make the complete development process outcome-blind.

Three analytical distinctions apply throughout. First, the performance of the final locked alarm is different from the performance of the development procedure estimated through resampling. Second, an overall dominant topic calculated across follow-up is a retrospective descriptor; prediction at a specific date may use only information available by that date. Third, a comparison anchored to a subsequently observed progression supports association and physiological interpretation; it is not, by itself, an operational prospective alarm.

3 Objectives and hierarchy

3.1 Primary evaluation objective

Estimate the sensitivity and specificity of the fixed final development alarm for first radiographic progression within 120 days after an eligible monthly decision point in the approved independent validation evaluation population. Both measures will be reported jointly, with confidence intervals, the underlying counts and the proportion of the intended monitoring period for which predictions can be evaluated. This is an estimation objective. No pass or fail criterion, superiority claim or equivalence margin is inferred from the available cohort size.

The primary target is performance at decision points with an administratively mature 120-day outcome horizon, defined without reference to whether progression occurred. Additional analyses address earlier clinical loss to follow-up and all eligible decision points with support for censoring adjustment. The distinction is specified in Section 10 and applies to development threshold selection as well as evaluation.

3.2 Secondary evaluation objectives

Estimate PPV, NPV, AUROC of the continuous rolling score, alarm episode burden, false-alarm burden, event-level detection and lead time. Describe the distribution and availability of the fixed digital states. Assess performance after excluding the recovered pre-cutoff records and in the strictly temporal subset whose recruitment provenance is confirmed.

3.3 Development and explanatory objectives

Reproduce all results affected by the clinical audit; quantify within-person signal changes; assess internal optimism and representation stability; evaluate incremental value over simple comparators; investigate missingness and heterogeneity; and assess molecular concordance and the operational implications of an alarm. These analyses are exploratory unless a specific hypothesis and its analysis were fixed before the relevant results were inspected. Results will not be selected for presentation according to statistical significance.

3.4 Analysis sequence

The required order is clinical reconciliation and data freeze, reconstruction of development results, completion of development-only methodological decisions and comparators, algorithm freeze, and evaluation in VAL-SET. No independent validation population (herein, “VAL-SET”) performance may be used to select a development variant.

4 Data sources and analysis populations

4.1 Source releases

The accompanying annex identifies four supplied workbooks by full filename, worksheet and SHA-256 checksum. They represent the historical development export, its audited counterpart, the historical matching export and the VAL-SET evaluation export. Original files will remain unmodified. A corrected analysis release will have a new version, checksum and signed change log.

The historical and audited exports contain the same 178 unique identifiers. As supplied, the historical file codes 62 PD events and 26 early progressors; the audited file codes 60 and 24 respectively. Four audited end dates occur after the 11 September 2025 nominal closure date stated in the provenance report. A filename, worksheet name or modification date is not evidence of a clinical cutoff.

VAL-SET contains 48 unique candidate participants and no identifier overlap with either 178-participant export. Its stored administrative nominal closure date is 3 September 2026. Forty-five reconciled start dates are later than 11 September 2025, two are earlier and one is on that date. Recruitment dates and the development input manifest are needed to establish the temporal and model-exposure classification.

The nominal closure dates were 11 September 2025 for the original development cohort and 3 September 2026 for VAL-SET. At each closure, centres were completing outstanding clinical records and closing assessments that had already been requested before the relevant nominal date. Records confirmed to belong to these pending requests were retained when received later. Data entered after the nominal date for an earlier assessment retain the actual assessment and event dates. Where the closing assessment itself occurred after the nominal date, its actual date is retained as a documented participant-specific exception. Membership of each analytical release

remained fixed, and completion was limited to those pending closure requests; newly recruited participants and successive later visits were not added to the same release.

The final release record will identify the affected participants, the pending request or source justification, the actual assessment date and the resulting analysis end date. Inclusion under this policy is based on the documented closure request, rather than on whether the assessment showed progression or on the alarm result. The nominal closure date, completion of outstanding records and final database freeze will be reported separately.

4.2 Development populations

Population	Membership rule	Intended use
D178 clinical population	The 178 original identifiers, with adjudicated clinical data and the approved original observation window, including documented closing-assessment exceptions	Clinical description, source reconciliation and analysis flow
D178 digital population	Members of D178 with the raw signal inputs needed for a specified analysis	Availability, digital summaries and reconstruction
D178 topic population	Members of D178 meeting the approved monitoring and topic-inference requirements	Topic descriptions and longitudinal states
D178 alarm population	Members of D178 with at least one valid score at a decision point defined in Sections 5 and 7	Alarm threshold derivation and internal evaluation
D178 matched population	Cases and controls newly selected using adjudicated data and the rules in Section 9	Event-anchored and within-person analyses
D178 molecular populations	Members of D178 meeting the modality-specific rules in Sections 18 and 19	ctDNA and molecular anchoring

Each population will have its own flow and denominator. A filled monthly topic cell, 30 days between two dates, and 30 days of usable digital monitoring are not interchangeable criteria. Participants who cannot generate an alarm remain in the clinical flow and availability reporting.

4.3 VAL-SET independent evaluation population

The principal VAL-SET population comprises all 48 candidates who satisfy the clinical eligibility criteria and for whom absence from relevant development inputs is confirmed. This includes recovered participants with earlier recruitment if they were unused in representation fitting, preprocessing estimation, grouping, selection and tuning. Absence from the original CRF export alone is insufficient evidence. Identifiers must be checked against the VQ-VAE source manifest where relevant, the HDO LDA input roster, all development signal exports and tuning records.

The complete population will be described as evaluation in participants not used for development. It will not be described as wholly recruited after the original cutoff if recovered participants are included. A required sensitivity analysis will exclude IDs 21003, 91021 and 31017. A strict temporal subset will be defined from actual recruitment dates after the approved original cutoff.

Any candidate with documented development exposure is excluded from the independent evaluation population and retained in the flow with the reason. An unresolved exposure history is reported as unresolved rather than assumed negative. The membership decision is made without reference to alarm performance. No participant is moved into D178 because of their evaluation outcome.

5 Time origins and construction of digital windows

5.1 Separate clinical and digital clocks

The clinical origin T0 is the verified date of first systemic treatment. PFS and the definition of early progression use this origin. Recruitment, consent, onboarding, the first signal and first treatment must be stored separately.

The digital origin D0 is the first usable signal date in the approved signal release, following the software specification for eligible inputs. The requested re-anchoring uses D0 for consecutive monthly digital blocks. D0 must be verified from signal timestamps.

The proposed standard is consecutive 30-day blocks beginning at D0. Block 1 contains calendar days D0 through D0 plus 29 days; its information becomes available only after that block is complete. Block m contains the corresponding next 30 days. The decision is made at the end of the final date of the completed block. For date-only outcomes, its decision_date is that final date and the event window is decision_date < PD_date <= decision_date + 120 days. The first calendar day after the completed block is therefore included. The implementation manifest must state timestamp timezone, day-boundary convention, daylight-saving handling and whether all source events are stored as dates or timestamps.

If the archived implementation used calendar months or a different boundary, the discrepancy must be resolved before rebuilding and freezing. The primary representation will not silently mix calendar months, four-week intervals and 30-day blocks. A choice to adopt the 30-day convention is recorded as an analytical specification change and applied identically to development and VAL-SET

No alarm is emitted before treatment starts or at or after first clinical progression, radiographic progression or death. Pretreatment digital days, where available, are handled according to an explicit rule in the algorithm manifest. Their use cannot be inferred from the Excel column names.

5.2 Clinical end and digital end

Clinical follow-up and digital availability end for different reasons. The analysis retains first radiographic PD, first clinical PD, death, last adequate disease assessment, withdrawal from digital monitoring, withdrawal from clinical follow-up and administrative cutoff as separate dates and reasons whenever available. Loss of a wearable signal does not automatically censor clinical follow-up, and a later clinical assessment does not create digital observations in an empty interval. For VAL-SET, derived durations must use the approved censored clinical end field and not the existing Tiempo field without checking its formula.

5.3 Topic inference and summarisation

All approved daily features, quality rules, masks, normalisation constants and the VQ-VAE and LDA objects are applied without refitting. Re-anchoring and recomputing posterior topic assignments are inference operations. The monthly posterior must be generated using the archived aggregation procedure, not an undocumented average of a different daily quantity. The manifest states whether a month is inferred from its day-type histogram or summarised from daily posterior vectors.

Each available six-topic probability vector must contain six finite values in the interval from zero to one and sum to one within a numerical tolerance of 0.000001. Missing vectors remain missing. A month fails eligibility if the locked minimum digital coverage rule is not met. A valid-day mask and monthly observed-day denominator are exported alongside each vector. Neither missing months nor discarded days are silently assigned a favourable state.

A 28-day (4 weeks minimum monitoring) eligibility rule, a 30-day block and a 90-day accumulation of rolling-adverse DCABCP burden for the alarm are distinct decisions. Their values are entered explicitly in the manifest; they are not inferred from which rule produces more evaluable VAL-SET patients.

6 Clinical and analytical outcomes

6.1 Progression free survival

PFS is elapsed time from T0 to the first documented radiographic or clinical progression, or death from any cause, whichever occurs first. Participants without an event are censored at the last adequate disease assessment within the approved participant observation window defined under section 4.1. A treatment-stop date or date of data entry is not automatically a PFS event or censoring date.

The source event fields require adjudication because the exports do not separately certify all death and progression components. An analysis using a PD-only field will be labelled time to documented PD unless the PFS definition has been verified. Death without prior PD must not be reclassified as radiographic PD.

6.2 Alarm target

The primary alarm target is first radiographic progression within the 120 days after a decision date, including the upper boundary and using the date convention in Section 5.1. The target event must be adjudicated using the imaging and clinical record; a CRF entry date is not substituted for the date of progression. Results will also be shown for the clinically broader composite of first progression or death as a prespecified sensitivity analysis if the component dates are complete.

Death before radiographic PD and first clinical progression without radiographic confirmation are recorded as separate terminal competing outcomes for the radiographic target. Both end the prediction risk period; scoring does not continue after clinical progression to seek a later scan-confirmed event. Neither is treated as an ordinary independent loss to follow-up. For this binary 120-day cumulative-incidence target, each is a known non-case of the radiographic endpoint; its frequency and alarm status are shown separately so that apparent reassurance or a false-positive label is not misinterpreted clinically. The composite sensitivity analysis counts both as events. Clinical adjudication must establish whether apparently separate clinical and imaging records describe the same progression before these rules are applied. Any change to the primary target requires an explicit amendment before threshold derivation and VAL-SET performance evaluation.

6.3 Early progression

Early PD is first documented PD before day 180 after T0. A PD exactly at day 180 is not early under this definition. A sensitivity analysis using PD on or before day 180 may quantify the effect of the boundary; it will not determine the main definition after results are seen.

6.4 Availability and lead time

Availability is the proportion of intended decisions at which the locked pipeline can produce a score. It is described both per patient and per monitored patient-month. Reasons for no score include insufficient initial history, inadequate current or prior coverage, invalid input, data transmission gaps and study termination.

Lead time is elapsed time from a qualifying alarm to first target progression. The primary event-level lead time uses the first alarm within the prespecified 120-day window before that event. Earlier alarms outside that window are not relabelled as true positives. Lead time is conditional on detected events and is reported together with the number and timing of undetected and operationally non-evaluable events.

7 Frozen primary alarm and development threshold

7.1 Components to freeze

The algorithm certificate records the feature dictionary, source device and vendor versions, valid-day rules, missingness masks, preprocessing constants, VQ-VAE weights and codebook, LDA fit and topic-word distributions, topic labels, favourable and unfavourable mapping, digital clock, monthly aggregation, window completeness rules, score calculation, decision cadence, target definition, horizon, threshold and comparison operator. It also records software versions, repository commit, random seeds and checksums of model objects.

The primary alarm uses six topics with the submitted mapping of 0 to 2 as favourable and 3 to 5 as unfavourable. Monthly pUF is the sum of posterior probabilities for topics 3, 4 and 5. AUC3 is the sum of pUF in the current and two preceding consecutive eligible monthly blocks. Its scale is zero to three. An alarm is positive when AUC3 is greater than or equal to the final development threshold. The primary cadence is one decision per completed 30-day block, with an accumulation of three blocks and a prediction horizon of 120 days.

A missing block within the three-block history prevents a primary score unless the documented reference implementation contains a different, explicit missing-data rule that is approved before freezing. The score is never renormalised over only the observed blocks without declaring a different algorithm. The release must verify whether this discrete sum matches the archived AUC implementation; a daily integral, trapezoidal calculation or duration-normalised mean is not substituted without a documented development-only specification change. Rolling signal area AUC3 must be distinguished from ROC area AUROC in labels and interpretation.

7.2 Threshold re estimation

During preparation of the revised analysis, the clinical records of the original 178 participants were audited within the approved development observation window. Source reconciliation corrected treatment-start dates, progression or censoring dates, and some event indicators. Monthly DCABP probabilities will therefore be regenerated on the corresponding corrected observation windows using the archived VQ-VAE and LDA objects, while retaining the six-topic representation and the favourable/unfavourable grouping. Because alarm-threshold estimation depends on the alignment of digital histories with subsequent outcomes, this outcome-informed development step must be repeated using the reconciled dates and event status. The threshold will be re-estimated in D178 under the score definition, accumulation window, prediction horizon and selection criterion specified in this SAP, without refitting the representation. The resulting complete alarm specification will be frozen before evaluation in VAL-SET and applied there without threshold adjustment or other model updating.

The final threshold is re-estimated exclusively in the adjudicated D178 alarm population after re-anchoring. The reference selection criterion is the maximum of Youden sensitivity plus specificity minus one, using the same point eligibility and outcome rules as the primary evaluation. All candidate thresholds at distinct observed scores, together with the all-positive and all-negative limits, are evaluated. If several thresholds tie within numerical tolerance, select the threshold with higher sensitivity; if still tied, select the lower numerical threshold. Record the candidate table and the selected value without rounding for execution.

If clinical censoring before 120 days is present within administratively mature points, threshold selection uses the same prespecified censoring-adjusted estimators described in Section 10. The complete-outcome table is retained as a transparent secondary description. The choice of estimator is not switched according to the selected threshold or VAL-SET result.

The submitted threshold of 1.61 is retained in the provenance record but is not copied into the new certificate unless it is the actual selected value. The exact development-derived threshold and its derivation date will be entered in the algorithm freeze certificate. The selected threshold is a component of a revised development system, even if the representation weights remain unchanged. Performance estimated in the same D178 data used for this selection is apparent performance.

7.3 Evaluation restrictions

VAL-SET is used to apply the locked objects and calculate prespecified evaluation summaries. No fitting, alternative topic grouping, threshold search, W or H selection, demographic normalisation learned from the VAL-SET population, or comparator selection is permitted in the primary evaluation. A transformation intrinsic to patient-specific deployment is allowed only if it is already specified and uses that participant's past data alone.

Any change after evaluation begins creates a new model version. Its performance in the same VAL-SET release is then an updated or exploratory analysis, and the first locked evaluation must remain reported. Further evaluation of the updated version requires a new independent test or a clearly stated limitation [3,4].

8 Reconstruction of the submitted manuscript after clinical audit

8.1 Clinical reconciliation

Create one participant-level adjudication record containing original and corrected values, the reason, source document, reviewer, review date and implications for the analysis. Review all changes in treatment origin, progression status or date, death, withdrawal and last assessment. Do not infer the clinical reason from the direction of a numerical change. Implement the participant-level adjudications recorded in the accompanying reconciliation annex, including correction of treatment origins, removal of baseline assessments previously classified as progression, reconciliation of radiographic and clinical event dates, and separation of specimen discontinuation from withdrawal from clinical or digital follow-up. Retain the original and corrected values, source references, adjudicator and adjudication date. Apply the documented closure-completion rules to assessments occurring after the nominal closure date.

Regenerate the participant flow and all derived intervals from calendar dates. Reconcile the initial manuscript counts, original export counts and audited export counts.

8.2 Rebuild dependent digital results

Re-infer the affected day types and topic posteriors under the unchanged approved representation, then regenerate all summaries that depend on the analysis window. The existing original and audited monthly matrices differ in 174 participants. A code and model manifest must establish whether the differences arise from re-anchoring, corrected clipping, software changes or other processing.

Overall dominant topic is derived with the same definition as the submitted pipeline, documented in executable code. The original manuscript describes the most frequently dominant daily state across the observation period. An unweighted mode of monthly labels or an argmax of a global average is not an automatic substitute. Ties are resolved using a fixed rule documented before evaluation, preferably the lowest topic index when the original code has no rule.

8.3 Results to regenerate

Output	Required reconstruction
Cohort description and flow	Eligibility, event counts, follow-up, withdrawal, digital availability and denominators
Figure 2 and corresponding supplementary tables	New case and control eligibility, matching, baseline and event-anchored windows, units, descriptive summaries and tests

Output	Required reconstruction
Figure 3	Topic assignment, interpretation summaries and distributions by tumour type and age using the audited observation window
Figure 4	Overall and early states, switches, retrospective descriptive survival plots
Figure 5	Monthly trajectories, alarm development, evaluation tables, illustrative cases and horizon-consistent lead times
Figure 6 and molecular tables	Any group-based genomic or transcriptomic comparison affected by a changed topic label, with updated denominators and any expanded eligible samples

Supplementary table numbering will be reconciled with the final manuscript layout. The analysis ledger links each old panel or table to its new source release, script and output. Original outputs remain archived; changes are explained without claiming that they are necessarily small.

9 Within person digital changes and renewed matching

9.1 Historical matching audit and replacement

The first version matching export contains 26 case sets, each with five control records, for 156 rows and 126 unique people: 26 cases and 100 unique controls.The export contains two former cases whose PD flags have changed after audit and non-uniform event-window bounds.

For the revised analysis, generate matching from D178 clinical data after adjudication. The random seed is fixed at 20261031 and recorded with the software version and sorted input roster. For each early-PD case, define its index time as days since T0 to PD. Select up to five distinct controls of the same tumour type who are alive, under clinical observation and free of the corresponding event at that index time. No control can be the case itself or appear twice in the same set. Controls may be reused across sets when the eligible pool is small, with all reuse reported. A participant who progresses later can be an at-risk control at an earlier index time; this risk-set definition avoids requiring future event-free survival as an eligibility condition.

This risk-set analysis is the primary revised event-anchored comparison. For comparability with the submitted analysis, repeat the contrast using the historical control restriction to participants not classified as early PD, with the same corrected dates and transparent handling of censoring before day 180. The choice of control pool is not adjusted to recover prior P values.

9.2 Windows and availability

The baseline window comprises treatment days 1 through 14, with treatment day 1 equal to T0. The pre-index window comprises the 14 calendar dates immediately preceding the index date, from index date minus 14 days through index date minus 1 day. Use the same index offset from the matched control's own T0. For the primary within-person comparison, both windows must be complete as calendar windows and their date ranges must not overlap. Cases whose windows overlap remain in the clinical and early-PD flow but do not contribute to that paired-window estimand.

Compute participant-level window summaries from the verified daily signals, not the historical matching table. The primary window summary is the median of valid daily measurements. Require at least eight valid days in each 14-day window as an explicit revision-analysis rule, with sensitivity requirements of at least five and at least ten days. These rules are applied to each signal separately and are distinct from the primary alarm's inherited coverage rules. No absent measurement is converted to zero.

9.3 Estimands and inference

For each signal, calculate the within-person change from baseline to pre-index and compare the case change with the mean of its matched controls' changes. Average the matched-set contrasts with equal case-set weight. Report changes in native units; percentage changes may be shown secondarily when baseline values are strictly positive and clinically interpretable.

Confidence intervals and tests must account for repeated use of controls and the matching procedure. The primary resampling unit is the unique participant in the original eligible cohort; eligibility, matching and window contrasts are repeated within each resample, preserving all observations and uses of a participant together. An equivalent prespecified crossed patient and matched-set model may be used if its assumptions and implementation are documented before execution. Treating 156 rows as independent is not allowed. Sensitivity analyses prohibit control reuse where sufficient controls exist and use one control per case.

Apply Benjamini-Hochberg adjustment across the primary signal interaction or change-contrast tests as one family. Report effect estimates, 95% intervals, exact denominators and both P and q values. Baseline group differences and absolute pre-index contrasts are descriptive supporting analyses. No signal is selected for the composite representation because it is significant in this block.

10 Evaluation of the fixed alarm in VAL-SET

10.1 Primary decision point population

An intended point is the end of a completed monthly block while the participant is alive and before first clinical or radiographic progression. A score-eligible point additionally has the required three-block history and signal quality. An administratively mature point has a complete 120-day potential calendar horizon before the fixed VAL-SET administrative cutoff, irrespective of the observed outcome. The primary decision-point analysis uses score-eligible, administratively mature points in the approved VAL-SET population.

For the primary decision-point analysis, 3 September 2026 remains the common administrative maturity reference, irrespective of participant-specific completion of pending closing assessments. Eligible decision dates must satisfy decision date + 120 days ≤ 3 September 2026; the latest such date is therefore 6 May 2026. A progression observed within the 120-day horizon is retained without requiring further follow-up after that progression. Participant-specific later closing assessments do not extend eligibility for decision dates in this primary analysis.Positive events observed within the horizon are retained. A patient lost clinically before the horizon without a known event has an unknown label, not a negative label.

The primary estimand gives equal weight to each eligible decision point, describing performance for a randomly selected evaluable monthly decision in this monitoring setting. Because patients with longer monitoring contribute more points, a sensitivity analysis gives equal total weight to each participant across their eligible points. Event-level analyses separately address detection of distinct progressions.

10.2 Censoring and competing events

For each point, classify the observed course as radiographic PD within 120 days, observed free of progression and death through 120 days, a terminal competing outcome within the horizon, or clinically censored before the horizon with no known outcome. Terminal competitors are defined in Section 6.2 and reported separately by type. The first three categories have known status for the radiographic cumulative-incidence target. Early clinical censoring requires adjustment or an explicitly conditional descriptive analysis.

If such censoring is absent, compute the primary operating characteristics directly. If present, use inverse probability of censoring weighting under a stated independent-censoring assumption within the analysis

population. Estimate G(u), survival free of clinical censoring at residual time u, with Kaplan-Meier; PD and terminal competitors end observation without counting as clinical censoring events. A radiographic PD at u within H = 120 days contributes as a case with weight 1/G(u-). A terminal competitor within H contributes as a non-case with weight 1/G(u-), while retaining its separate clinical category. A participant observed free of either outcome through H contributes as a non-case with weight 1/G(H). A point censored before an outcome or H has no directly assigned binary label. Sensitivity, specificity, PPV and NPV are calculated from the resulting weighted two-by-two table; time-dependent AUROC uses the same case and non-case definitions. An outcome and censoring date tied at day resolution are adjudicated from source chronology, with a documented event taking precedence when it was observed that day. The implementation is reviewed before evaluation, and G is re-estimated in every patient bootstrap [6,8].

Report the distribution of weights, effective sample size and outcome counts. If the required censoring survival probability is zero or the estimate cannot be supported, the corresponding adjusted metric is not estimable. A weight cap at 10 is a prespecified sensitivity analysis, with uncapped estimates shown where finite. Complete-outcome tables are reported separately and are not presented as unbiased estimates for all intended points when early censoring is informative.

An additional analysis uses all score-eligible decision points with time-dependent censoring-adjusted performance estimation, provided the horizon has support. It is explicitly secondary because VAL-SET is small and administrative censoring is strongly related to entry time. No censoring method is used to claim observed information beyond the available follow-up [5,6].

10.3 Performance and uncertainty

Report sensitivity, specificity, PPV, NPV and their denominators at the locked threshold. Report AUROC for the continuous AUC3 score. Precision-recall summaries can be shown descriptively with the observed event proportion. The score threshold is not reselected for VAL-SET, and pUF is not interpreted as a probability of progression.

Use 2000 bootstrap resamples of unique participants, preserving each person's entire trajectory and all repeated decision points. The primary intervals are percentile 95% intervals. Report the number of bootstrap replicates in which each metric is undefined, and do not silently replace an undefined value. If fewer than 90% of resamples yield a finite metric, present the point estimate with an explicit unstable or non-estimable interval and the raw counts. A binomial interval treating all repeated months as independent is not used.

Tumour-specific VAL-SET results are descriptive counts with intervals when estimable. No separate tumour threshold, multivariable refit or confirmatory interaction claim is supported by this small test population.

10.4 Alarm episodes and event level detection

The primary operating characteristics treat each scheduled decision separately. For operational burden, a new episode begins on the first positive decision or on a negative-to-positive transition. Consecutive positives are one episode. A missing interval does not prove resolution; a later positive after a gap is labelled a resumed or uncertain episode and reported separately. A 30-day refractory rule is a prespecified alternative for finer cadences.

Report episodes per 100 patient-months, participants with at least one alarm, repeated positive months, and false-positive decisions with observed event-free horizons. Report the fraction of distinct PD events with at least one positive decision during the preceding 120 days, conditional on at least one score opportunity, and separately relative to all clinical PD events. The latter includes events before the three-block run-in and events with no score as not detected by the system, with reasons shown.

Summarise event-level lead time with individual values and a median if meaningful. An alarm ten months before PD remains outside the four-month target for that decision. An apparent false positive is not converted to a true positive by extending the horizon after observing the outcome.

11 Topic distributions and survival analyses

11.1 Descriptive distributions

Describe the six fixed topics and favourable/unfavourable states in D178 and VAL-SET using patient and patient-month denominators separately. Show distributions by tumour type and age using development-defined age boundaries. The VAL-SET age cut points are not re-estimated. Report missing and unclassifiable observations and the number of patients contributing to each display.

The overall dominant topic and the original month-1-to-overall transition plots will be reconstructed to maintain correspondence with the submitted figures. Their association with PFS is explicitly retrospective because the grouping uses information acquired during follow-up. They are not the primary evidence of prospective prediction.

11.2 Time varying analysis

Fit a start-stop Cox model in D178 with continuous pUF as the primary time-varying digital covariate and the binary favourable/unfavourable state as a sensitivity analysis. Entry occurs when the first valid monthly state becomes observable, with time measured since T0 and delayed entry recorded. Update only at the end of each completed block; a state is not backdated to the start of its source month. Carry the last valid state only until the next scheduled monthly decision. If that next state is unavailable, the following interval has unknown digital state and is excluded from the complete-state association analysis until observation resumes; report excluded intervals and a secondary model with an explicit unknown-state category. Do not interpolate using a future state. Use participant-clustered uncertainty and the adjustment strategy in Section 16. Check proportional hazards for fixed covariates; report time interactions or stratification if required. VAL-SET receives descriptive unadjusted time-varying estimates only if the data support estimation.

Reconstruction of the original overall-state Kaplan-Meier figures will be clearly labelled as different estimands. Future-defined overall labels will not be treated as baseline covariates in a causal or prospective prognostic interpretation.

12 Internal evaluation of the development procedure

The independent VAL-SET evaluation assesses the final frozen system. Internal resampling assesses optimism and instability in the development process. Neither substitutes for the other [3]. The primary internal approach is patient-level nested five-fold cross-validation repeated ten times, with fixed seeds derived from 20261031 and stratification by tumour type and event status where feasible. All observations of one patient remain in one fold.

Within each outer training set, repeat the relevant HDO-derived preprocessing estimates, LDA fitting, outcome-informed favourable/unfavourable mapping and threshold selection. The external VQ-VAE object may remain fixed only when independence of its pretraining data from the held-out participants is verified. The number of topics remains six for evaluation of the specified six-topic algorithm; a claim about the original model-selection process requires the archived candidate choices to be reproduced inside the inner loop.

The inner loop performs any tuning allowed by the prespecified candidate list. W equals three blocks and H equals 120 days for evaluation of the revised fixed architecture. If the objective is to estimate optimism of the historical search over metric, W and H, recover and repeat that exact search inside training folds and label it separately. Do not claim that fixing a historically selected choice corrects optimism from its original selection.

Full-process evaluation requires executable LDA refitting and an archived or prospectively specified training-only rule for assigning favourable and unfavourable topics. If either is unavailable, report evaluation as conditional on the fixed representation or historical grouping, as applicable; it will not be labelled full nested validation of the complete procedure. A newly specified grouping rule evaluates a revised development procedure, not an exact

reconstruction of the historical choice. Topic alignment for reporting uses training-set distributions and not held-out clinical outcomes.

Leave-one-centre-out evaluation is a complementary analysis where source centre identifiers and sufficient event counts are available. Each held-out centre is excluded from the relevant learning stages. Report centre-specific counts and uncertainty, and separate site and tumour effects when their distributions differ. An uninformative fold is reported as such. Leave-one-patient-out is not mandatory in addition to nested cross-validation and is not used as a substitute for grouped splitting.

The fold-specific fitted objects are archived for evaluation only. The final D178 object used in VAL-SET remains the separately approved frozen model. No VAL-SET participant is used in selecting the internal validation method or its hyperparameters.

13 LDA stability and fixed representation assessment

Assess LDA stability in D178 using 200 participant bootstrap samples and at least 20 predetermined random initialisations. Keep the external codebook fixed. Align refitted topics to the reference topic-word distributions by minimum-cost matching using Jensen-Shannon distance; the matching uses no clinical labels. Report topic distance, prevalence, patient assignment agreement, posterior-probability agreement and the stability of the favourable/unfavourable mapping when that mapping is rederived within training data.

Assess sensitivity to the number of topics using a prespecified small grid of four through eight topics in D178 only. Report model fit and interpretation across the grid without replacing the primary six-topic model according to VAL-SET results. Repeated seeds and bootstrap fits estimate instability, not additional independent patients.

In VAL-SET, apply the fixed representation and report topic prevalence, posterior uncertainty and unfamiliar day-type frequency relative to development. This is a transfer assessment. Refitting LDA on D178 plus VAL-SET, renaming topics using VAL-SET outcomes or choosing a different topic count from VAL-SET would constitute model updating and is excluded from the primary evaluation.

14 Cadence and temporal granularity

The primary algorithm remains the locked monthly three-block score. Prespecified secondary analyses distinguish the frequency of decision-making from the mathematical definition of the score. Updating a fixed 90-day summary daily or weekly is a cadence change; replacing a sum of monthly posteriors with an average of daily posteriors is also a score change and must be labelled accordingly.

Using raw timestamped signals in D178, construct daily, weekly and fortnightly decision schedules and a trailing 90-day unfavourable-burden summary using only past data. Document coverage requirements and the handling of missing days. A daily probability average is not assumed to equal the original three-month sum. Derive any required thresholds exclusively in D178 under the same horizon and operational rules, then freeze each named secondary variant before VAL-SET evaluation.

Report score agreement, threshold-crossing agreement, event-level lead time, false-alarm episodes and availability for each cadence. Compare variants on common patients and aligned eligible opportunities as well as their full operational coverage. Dense overlapping windows do not increase the number of independent subjects. If raw inputs or timing provenance are unavailable, this block is reported as not executed rather than simulated from coarse monthly values.

15 Simple comparators and incremental predictive value

15.1 Prespecified comparators

The main digital comparators are a steps-only score equal to minus the median daily steps in the completed preceding 30-day window, and a heart-rate-only score equal to the median daily heart-rate summary in that window. Higher values therefore represent decreasing activity or higher heart rate. Secondary, separately labelled comparators use minus the change in steps and plus the change in heart rate from the patient's first 30 treatment days, only once the entire baseline window is observable. Standardisation parameters and threshold choices are learned only in D178. A secondary two-signal comparator combines the two current-window scores using a penalised logistic model. The daily heart-rate summary and valid-day requirements are fixed in the feature dictionary. ECG heart-rate variability and Garmin stress-derived estimates are not silently substituted for that variable.

The clinical comparator uses tumour type, age and baseline ECOG, with a prespecified compact extension incorporating treatment class when identifiable. A combined model adds the fixed digital burden to the same clinical variables. Additional baseline blood variables may be assessed in a separately named exploratory extension, using their verified units. A current ECOG comparator requires timestamped serial ECOG; baseline ECOG is not presented as an assessment made on the alarm date.

15.2 Fair comparisons

Use the same prediction horizon, clinical outcome definition and time restriction for each comparison. Compare paired performance on the intersection of evaluable points, and separately report each method's availability in the intended population. A comparator requiring fewer monitoring days may have greater availability; that difference is part of its operational value and is not hidden by restricting every analysis to the composite model's successful points.

Select thresholds using the same development criterion as the primary alarm. A secondary comparison matches development sensitivity and compares specificity, or matches alarm burden and compares event detection. These operating points are fixed before VAL-SET evaluation. Report paired differences with participant bootstrap intervals. Do not assert superiority from different apparent AUCs calculated in different populations.

15.3 Risk calibration layer

The topic probability pUF and the rolling score are not probabilities of progression. To address calibration explicitly, fit a separate secondary risk model in D178 mapping the continuous AUC3 score to 120-day risk, using a compact penalised logistic specification and the censoring strategy in Section 10. Select the penalty inside training resampling and freeze coefficients and any intercept. A clinical-plus-digital risk model is assessed under the same restrictions.

Evaluate the locked risk layer in VAL-SET with observed-to-expected risk, Brier score and calibration summaries only where censoring support and event counts allow. A smoothed calibration curve or slope is not interpreted as reliable with only a few independent events. No VAL-SET recalibration is part of the primary evaluation. Presenting a secondary risk layer does not change the original threshold alarm [4,5].

16 Tumour type treatment and concomitant medication

16.1 Heterogeneity

Describe clinical and digital data by breast, lung and colorectal cancer, including treatment, metastatic distribution and symptom burden when available. Report topic prevalence, score distribution, availability and alarm

performance by tumour in D178, with counts and participant-level intervals. Interactions between score and tumour type are exploratory and assessed jointly before interpreting individual contrasts. VAL-SET subgroup results are descriptive because the number of events is small.

Leave-one-tumour-type-out analyses may assess transfer to the held-out tumour category, but must exclude that category from the appropriate learning and threshold-selection stages. They are not geographic external validation. No tumour-specific VAL-SET threshold is fitted.

16.2 Multivariable associations

The core D178 survival model contains the binary digital state for the survival analysis, or continuous pUF for the primary time-varying analysis, age per ten years and baseline ECOG, with baseline hazards stratified by tumour type. The time-varying binary state is a sensitivity specification; digital measures are not interchanged according to significance. Report event counts, parameter degrees of freedom, effect estimates and intervals. Avoid automatic stepwise or univariate P-value selection.

A treatment-adjusted extension uses the prespecified clinically coherent treatment groups in the audited release. If tumour and treatment are nearly aliased, report that structure and use a parsimonious within-tumour or treatment-stratified analysis rather than an uninterpretable coefficient. Penalisation can be used to stabilise a prespecified larger model, with its penalty selected inside development resampling and its estimates identified as penalised. No large multivariable model is fitted to the eight VAL-SET events.

Each multivariable model will use participants with complete data for all variables required by that model. The number of participants and events included, and the number excluded because of missing covariates, will be reported for each model. Missing a covariate will not exclude a participant from other analyses for which the required data are available.

16.3 Medications

Opioids and systemic corticosteroids are the principal concomitant exposures; beta blockers are a prespecified cardiac-signal sensitivity covariate. Baseline exposure means use known at or before T0 under a documented look-back window. Longitudinal exposure requires start and stop dates or visit-specific status available at the prediction time. Binary variables indicating any use during later follow-up, hospitalisation or premedication are not automatically baseline confounders.

Analyse baseline opioid and corticosteroid exposure jointly with the core covariates when estimable. Separate prophylactic steroid premedication from symptom-directed, chronic and hospital-associated steroids when the clinical record allows. Use timestamped exposure as a lagged time-varying covariate in a secondary model; if dates are unavailable, restrict inference to baseline or descriptive associations and state the limitation. Do not adjust a prospective score using medication prescribed after the decision point.

The adjusted analyses assess whether associations persist conditional on measured factors. They do not establish that the signal is independent of all disease burden, treatment toxicity or unmeasured confounding, and they do not identify a causal host-state mechanism. Missing drug status remains unknown rather than no exposure.

17 Missing data and pretraining domain

17.1 Missingness description and informative loss

Missingness will be described separately for each digital modality, distinguishing absent measurements, invalid measurements, transmission or device problems, hospitalisation, patient choice, cessation of digital monitoring, and administrative termination. The date and scope of each discontinuation will be adjudicated from the clinical and study records. Discontinuation of blood sampling alone will not be classified as withdrawal from clinical visits or digital monitoring. Days before modality activation, after the relevant authorised observation period, or after the clinical event will not be included as expected monitoring days.

To assess whether missing digital observations are associated with impending progression, all adjudicated D178 progression events with an eligible observation window will be considered, without restriction to early progression. For each event, the 14 calendar dates immediately preceding progression will be compared with the corresponding elapsed-treatment-time window in controls selected using the risk-set principles in Section 9.1. Controls must be alive, progression-free and under clinical observation at the index time; they may progress later. Control eligibility will not require future event-free survival or a minimum number of valid digital measurements.

For each modality and participant, report the number of days with an expected measurement, the number without a valid measurement, and the proportion missing. A window in which every expected measurement is missing remains informative for this analysis and will not be excluded by the valid-day threshold used for signal summaries. Windows with no expected monitoring days will be identified separately. Compare participant-level missingness proportions between cases and controls, with uncertainty accounting for matching and repeated use of participants. The analysis will distinguish reduced availability before progression from structural cessation of monitoring.

For each decision point, derive past-only completeness over the preceding 30 and 90 days, the length of the current gap and whether monitoring stopped. Compare subsequent progression and death according to these indicators, using the same horizon and clinical follow-up rules as the alarm. A missingness-only comparator is developed in D178 and frozen before any VAL-SET evaluation. An indicator of eventual withdrawal is not used before the withdrawal has occurred.

Report alarm performance and availability by prespecified coverage bands of below 50%, 50% to below 80%, and at least 80%, while retaining the algorithm's own minimum-coverage rule. If a band has no score-eligible points, report zero availability rather than an invented performance value. Standardise or adjust for observed differences in tumour, treatment and baseline status where the data permit. Associations between missingness and outcome are interpreted as evidence about observation processes, not proof of a specific reason for dropout.

17.2 Clinical outcomes after discontinuation of digital monitoring

To assess whether cessation of digital monitoring is associated with subsequent clinical deterioration, clinical outcomes will be sought for D178 participants with confirmed discontinuation before progression, including outcomes documented during routine care after digital study participation ended, where their use is permitted by the applicable consent and study authorisation. The date and reason for discontinuation, last known disease status, subsequent progression, death, and last clinical contact will be recorded separately. Administrative closure and cessation caused by an already documented progression will not be classified as pre-event withdrawal.

Comparisons will begin at the confirmed discontinuation date. For each participant who discontinued, the comparison group will comprise participants of the same tumour type who remained under observation, alive and progression-free at the corresponding time since treatment initiation. Controls may subsequently discontinue monitoring. Clinical outcomes in both groups will be assessed using the same ascertainment rules and a

documented observation limit. Subsequent progression and death will be reported with time-to-event methods that account for censoring; death before documented progression will be distinguished from progression. Participants will not be classified at treatment initiation according to whether they subsequently withdrew.

The number of eligible discontinuations, successful and unsuccessful outcome retrievals, available follow-up and observed events will be reported. If comparable outcome ascertainment cannot be established for the comparison group, the recovered outcomes will be presented descriptively rather than as an unbiased comparison of progression risk. These supplementary records will be used to investigate informative observation loss. They will not automatically extend the primary digital observation window, replace the primary endpoint record, or enter alarm fitting or independent validation. Source corrections concerning events that occurred within the approved primary observation period will continue to be handled through the clinical adjudication process.

17.3 Perturbation analyses

Apply random removal of 10%, 20%, 30% and 40% of otherwise observed days in D178, using fixed random seeds, and a separate experiment removing contiguous gaps of 3, 7 and 14 days. Re-run inference with the same model and missingness rules. Report posterior agreement, dominant-state agreement, threshold-crossing agreement, score availability and performance changes. Artificial random deletion assesses numerical sensitivity; it does not establish that real illness-associated missingness is ignorable.

17.4 External pretraining corpus

Archive the exact source cohorts, participant counts, inclusion dates, input variables and overlap checks for the VQ-VAE pretraining corpus. Verify the original statement that training was unsupervised; psychiatric diagnostic labels must not be described as an optimisation target unless they actually were used. The six HDO LDA topics are distinguished from the external 256-type codebook.

Assess shifts between pretraining data, D178 and VAL-SET using feature distributions, missingness patterns, codebook usage, reconstruction or quantisation error where defined, and topic posterior uncertainty. Use common variable definitions and fixed reference scales.

Development-only ablations compare the fixed pretrained representation with simple feature-based models and, if feasible with sufficient data, an oncology-only representation fitted within each training fold. Any candidate replacement is evaluated as a separate model. A positive masking experiment or an unsupervised objective does not by itself rule out domain-related bias. No claim is made that using a psychiatric-dominated corpus either invalidates or guarantees transfer to oncology without these empirical checks.

18 Expanded ctDNA analysis

18.1 Eligible molecular populations

Include all D178 participants within the approved historical clinical window who have a documented tumour alteration covered by the plasma assay, at least one baseline or early-treatment plasma measurement and at least one subsequent technically valid measurement, with reliable draw dates and assay quality information. Enumerate screened participants, assay failures, insufficient sample volume, untraceable alterations and missing specimens. Inclusion cannot depend on visually convincing digital and molecular concordance.

The prespecified strata are EGFR-mutant lung cancers, with treatment class identified including EGFR-directed therapy, and KRAS-mutant lung cancers treated with any treatment regime. The supplied spreadsheets do not establish the number of molecularly eligible participants; these counts remain to be completed from the assay inventory.

The rationale is to assess the digital-molecular relationship in different molecular and therapeutic settings. EGFR or KRAS status does not establish a unique clonal driver in every tumour, guarantee shedding, or ensure that acquired resistance will retain the same detectable alteration. Chemotherapy and immunotherapy also exert selection pressure. Document target-specific treatment, resistance alterations and assay coverage where known rather than assuming the same clone explains every rebound.

18.2 Measurements and linkage

Use the quantitative plasma measure supplied by the validated assay, its units, limit of detection, limit of quantification and QC flags. Non-detectable values are left-censored; do not convert them to biological zero without qualification. Show the original detection status and an appropriate transformed quantitative axis for plots. Any numerical substitution for below-detection values is prespecified and tested in sensitivity analyses.

At each blood draw, link the digital summary available strictly before or at that draw, using a prespecified trailing window and no interpolation from future signal. The primary descriptive linkage uses the preceding 30-day pUF summary, with the locked three-block burden shown where available. Blood sampling and imaging dates remain visible. Molecular data are not used to define the six topics, their favourable/unfavourable grouping or the alarm threshold.

18.3 Analyses

Display individual trajectories for all eligible participants or provide a complete supplementary series, together with a cohort-level summary. The primary molecular summary is within-person directional concordance between changes in the digital measure and the measured ctDNA across consecutive scheduled specimens, with no substitution of later unscheduled draws to improve agreement. Report the number of participants and paired intervals, and account for repeated intervals per participant.

Secondary summaries include within-person associations between digital burden and log-transformed ctDNA, time between first observed molecular re-emergence and digital alarm, and time to radiographic PD. Lead-lag analyses are restricted to a prespecified set of lags and shown in full. Sampling intervals limit the precision of temporal ordering: a signal observed earlier than the next blood draw does not prove that it preceded the biological rise in ctDNA.

The molecular team may know the mutation and treatment required for assay interpretation. Where practicable, assay QC and eligibility are finalised without digital trajectories, and digital inference is performed without using ctDNA results. Record actual blinding rather than asserting it retrospectively. Analyses are exploratory biological support, with tumour and treatment strata retained. They are not external validation of the clinical alarm.

19 Other molecular anchoring and updated Figure 6

Recompute analyses that depend on corrected overall topic assignments using the existing approved tumour DNA and PBMC RNA-seq processing, retaining archived QC and normalisation procedures. Additional samples sequenced for the revision may be included when their collection date and participant eligibility meet the frozen population definition. List all assay-eligible participants and reasons for exclusion before testing topic associations.

Specify separately the genomic, transcriptomic and deconvolution populations. Adjust for tumour type, age and technical batch where relevant and identifiable. The relation between a baseline molecular measurement and an overall state derived from later follow-up is a retrospective association; it is not a baseline predictor unless separately evaluated with an appropriate fixed target.

Apply false discovery rate control within each declared molecular testing family. Gene-level, pathway-level and cell-composition analyses have separate reported denominators and families. Do not select a pathway from one assay and call the association confirmed in the same samples without identifying that selection. The biological

analyses provide plausibility and hypothesis generation; they do not remove outcome-informed selection in the alarm development.

20 Clinical pathway simulation and decision analysis

The prespecified intended response to an alarm is an additional clinical review, with earlier imaging or blood testing considered according to clinical assessment. The simulation does not replace scheduled surveillance, postpone care after a negative alarm, or assume a treatment change based solely on digital data. It estimates workload and observed event capture under explicit assumptions.

Use the locked primary alarm episodes and observed clinical timeline. Simulate review delays of 7 and 14 days and a 30-day refractory interval after an alarm-triggered review. Report reviews and potential extra tests per 100 patient-months, participants contacted, repeated contacts, PD events with a preceding alert, undetected events, and the observed time from alert to scheduled imaging. Without actual appointment or scan dates, label the corresponding measures as scenario calculations rather than observed comparisons with standard surveillance.

Show the consequences of false positives, false negatives and missing scores. Do not equate a high NPV with evidence that a scheduled visit can safely be cancelled. Do not estimate survival benefit or therapeutic benefit without interventional data.

Decision-curve analysis may be undertaken for the frozen secondary risk model over prespecified risk thresholds from 5% to 30%, with broader thresholds shown only as labelled sensitivity scenarios. The net-benefit calculation and assumed relative harms must match the proposed action, such as additional review. For the binary primary alarm, evaluate its fixed strategy under explicitly stated utility ratios; its score threshold is not a clinical risk probability. Compare with review-all and review-none where appropriate. Report censoring-adjusted estimates and uncertainty only when the data support them [7].

21 General statistical principles and multiplicity

Analyses emphasise estimates, uncertainty and availability. Report continuous variables with median and interquartile range or mean and standard deviation as appropriate, categorical variables with numerator and denominator, and time-to-event estimates with explicit event and censoring definitions. Do not infer a centre, tumour type, medication status or unit from an identifier pattern when a source field is absent.

The primary evaluation reports sensitivity and specificity jointly without a dichotomous validation-success declaration. PPV and NPV are interpreted in the context of the event frequency and decision-point population. Differences between development and VAL-SET may reflect optimism, case mix, treatment, follow-up and availability; they are not attributed to a single cause without evidence.

Use two-sided nominal 95% intervals and P values where a test is justified. Apply Benjamini-Hochberg adjustment within the prespecified within-person signal family, topic-distribution exploratory family, and each molecular family. Treat heterogeneity interactions, cadence variants and alternative models as exploratory comparisons; report all prespecified variants rather than selecting the most favourable one. A significant result in one subgroup and a non-significant result in another is not an interaction test.

The 178 and 48 available candidates determine the current sample size. No retrospective power calculation is used to claim adequacy. Report the number of distinct events and the precision of each estimate. Evaluation precision will be particularly limited by the small number of VAL-SET events with an opportunity for the three-month alarm. Additional months from the same participant do not substitute for additional independent participants [5].

Random seeds, convergence criteria, resampling failures and any separation or singular fits are retained. If a prespecified model is not estimable, report the reason and the simpler descriptive result; do not silently replace it with a model chosen for significance. All rounding is for presentation only; execution uses full stored precision.

22 Data freeze execution and reproducibility

22.1 Separate dates and approvals

Record the clinical data cutoff, the clinical audit completion date, the date a file release is frozen, the algorithm freeze date and the first VAL-SET performance evaluation date separately. The SAP date of 31 October 2026 is none of those dates unless the corresponding action actually occurs then. A file checksum establishes byte identity; it does not establish clinical correctness or prior blinding.

The development clinical release requires resolution of the cutoff discrepancies, endpoint definitions, reasons for corrected status and dates, units and clock semantics. The VAL-SET release requires its durations derived from the adjudicated observation windows, recruitment and recovered-record provenance, non-exposure checks and endpoint review. The algorithm freeze requires the complete certificate in Appendix B and reproducibility from the approved development release. These are concrete prerequisites, not a certification that the supplied workbooks are already final.

22.2 Access and execution ledger

Before approving this SAP version on November 2nd, the investigators had reviewed the original D178 manuscript analyses and clinical data used for the D178 audit and the assessment of candidate VAL-SET participants. The investigators confirm that the signed access and execution ledger accurately records the personnel, dataset versions, analyses and results reviewed before approval, together with the actual dates or documented date intervals. This includes any prior inspection of topic probabilities or pUF summaries, threshold searches, model fitting and alarm-performance assessments.

Clinical adjudication is distinguished from inspection of digital model outputs and their relationship with clinical outcomes. Analyses already performed or initiated, and methodological decisions informed by their results, are identified in the ledger; this SAP does not retrospectively prespecify them. The ledger also identifies the evaluations and decisions still pending at approval. The algorithm-freeze date and the first VAL-SET performance-evaluation date are recorded separately, irrespective of whether they precede or follow SAP approval.

For each analysis block record the input release and checksum, code commit, model object, parameters, seed, executor, execution date, result-inspection date and status. Mark analyses already completed before SAP approval as retrospective or previously initiated. Produce a manifest linking every table, figure and rebuttal claim to its executable source. Validation scripts are reviewed on development data or synthetic test cases before joining VAL-SET predictions to outcomes.

22.3 Software and quality review

Use a versioned Python environment with recorded package versions. Verify participant joins, date arithmetic, block boundaries, event-window labels, missingness handling, model identity and sample counts. Particular checks include events at day 180, events at the 120-day horizon boundary, events before the first score, a missing middle block, repeated controls, censored predictions and post-cutoff source dates.

Reproduce the primary output from the archived release and environment. The signed provenance record, source exports, corrected releases, change log and code history are retained without overwriting earlier versions.

23 Amendments and reporting

Each amendment states its date, reason, affected populations or algorithms, information already inspected and whether it precedes or follows VAL-SET performance access. Corrections that restore source truth are

distinguished from analytic choices. An event outside the approved observation window is not added to the release unless it meets the documented closing-assessment policy or is covered by a dated amendment.

The revised manuscript will identify the recovered-record sensitivity, the distinction between retrospective overall states and prospective information, operational coverage and uncertain estimates.

Appendix A Analysis blocks and reviewer coverage

Block	Main deliverable	Reviewer comments addressed
A01 Clinical reconciliation and reconstruction	Audited flow, change log and rebuilt submitted analyses	R1 C and D; R4 2 and 6
A02 Within-person changes and matching	Paired changes and corrected window comparisons	R2 comments on signals; R1 D
A03 Fixed alarm evaluation in VAL-SET	Operating characteristics, coverage and uncertainty	R3 2 and 5; R4 2, 3 and 5
A04 Topic and survival analyses	Fixed-state distributions and time-varying PFS	R1 D and E; R3 3 and 8
A05 Internal evaluation and LDA stability	Patient-level nested evaluation and topic stability	R1 D and E; R3 2 and 5
A06 Cadence and granular analysis	Cadence variants and alarm episodes	R2 comments on monthly summaries
A07 Simple comparators and calibration	Steps, heart rate, clinical and combined models	R3 1; R4 1, 5 and 7
A08 Missingness and availability	Reasons, missingness-only comparator and perturbation	R3 4 and 9
A09 Pretraining domain assessment	Corpus provenance and transfer diagnostics	R3 6; R4 7
A10 Tumour treatment and medications	Heterogeneity and adjusted associations	R1 D; R3 3, 7 and 8; R4 3 and 4
A11 Molecular expansion	Complete EGFR and KRAS series and updated Figure 6	R4 2; R2 molecular comments through revised presentation
A12 Clinical pathway simulation	Review burden, detection opportunities and decision analysis	R1 B, E and F; R2 clinical utility; R3 3; R4 5

The mapping is a response-planning aid. It does not imply that each block has been executed or that a single analysis fully resolves a reviewer concern.

Appendix B Algorithm freeze certificate

Complete and sign this certificate before the first performance evaluation of the approved VAL-SET release. If an item differs from the submitted method, record the reason and development-only evidence supporting the change.

Required item	Value or record to attach
Approved D178 clinical release and checksum	To be entered after adjudication
Approved VAL-SET release and membership list	To be entered after provenance checks
Approved original and VAL-SET nominal closure dates and participant observation-window exceptions	To be confirmed from source records
Raw signal release and D0 verification	Release, fields, timezone and first usable timestamp rule
Feature dictionary and preprocessing	Version and checksum
VQ-VAE weights and codebook	File identifiers, checksums and no-overlap evidence
LDA fit and topic distributions	File identifiers and checksums
Number of topics and grouping	Six; favourable 0 to 2; unfavourable 3 to 5
Monthly inference and partial-window rule	Exact implementation and approved coverage requirements
Primary accumulation and horizon	Three completed 30-day blocks; 120 days after decision
Threshold and operator	Exact development-derived value; greater than or equal to
Censoring and competing-event implementation	Reviewed code version and estimator
Primary and secondary evaluation populations	Patient and point manifests
Secondary comparators and cadence variants	Complete objects and parameters before VAL-SET performance evaluation
Software code and seeds	Repository commit, environment and seed manifest
Outcome access declaration	Named users, dates, prior analyses and result-inspection history
Freeze approval	Miguel Quintela-Fandiño; Pablo Martínez Olmos; Antonio Artés; Leire Paz; Leonardo Garma; actual dates and signatures

References

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